

INFRAESTRUCTURA 2 — VPN SITE-TO-SITE

Cisco C2691 + FortiGate + ISP + Web Server

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Seguridad En Redes

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1. Propósito

Implementar una infraestructura de comunicación segura entre una red de usuarios y una red de servidor mediante un túnel VPN Site-to-Site entre un router Cisco C2691 y un FortiGate. La práctica incluye direccionamiento IP, DHCP, NAT, Web Server HTTPS e ISP.

2. Objetivos

- Comunicar el Usuario con el Servidor a través del enlace VPN.
- Comprobar que la comunicación solo fluye cuando el enlace VPN está activo.
- Configurar R1 con red, NAT y VPN Site-to-Site.
- Configurar FGT-2 por GUI con red, NAT y VPN Site-to-Site.
- Implementar Web Server HTTPS en 10.229.3.128/28.
- Configurar Usuarios 10.229.3.0/25 con VLAN 10 y DHCP.

3. Topología

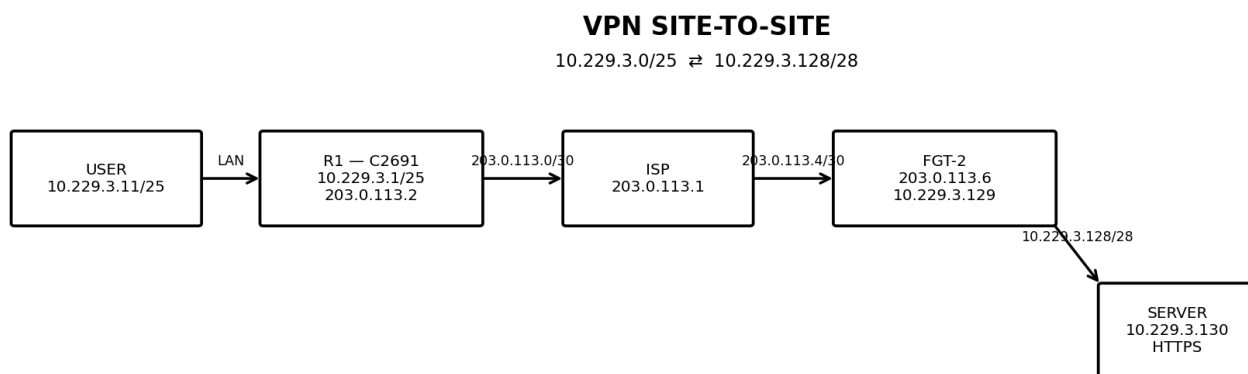
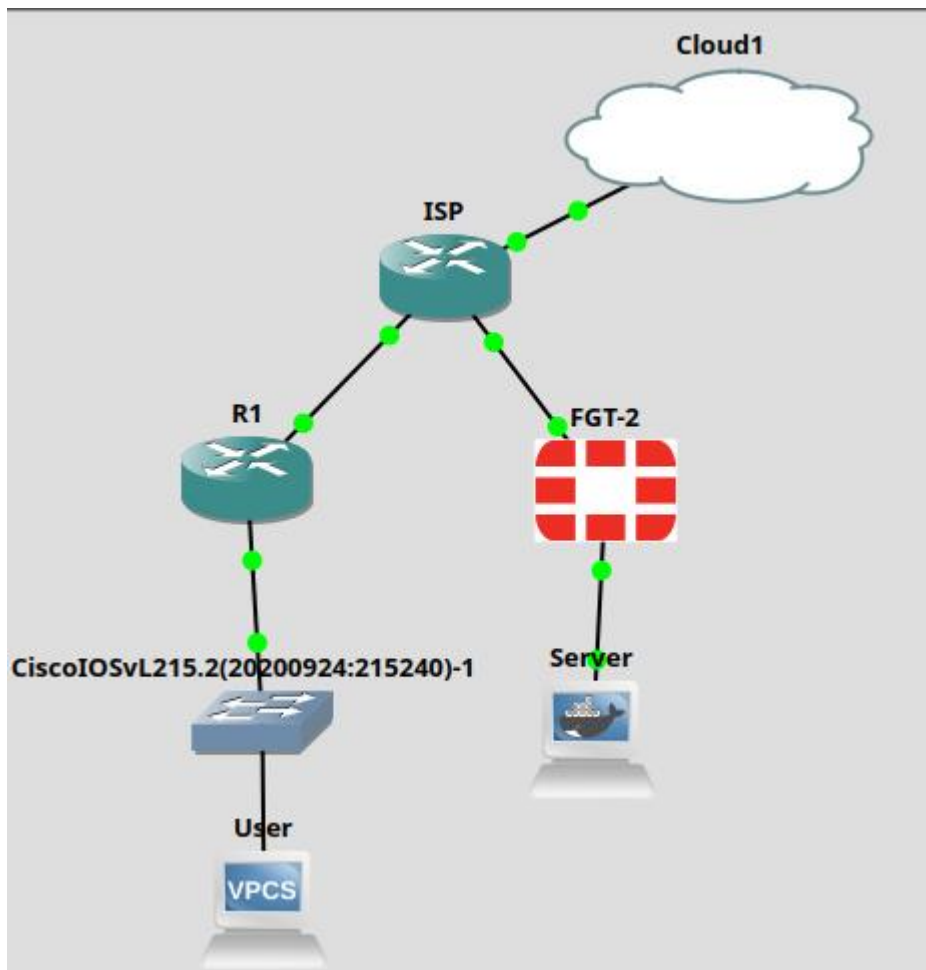


Figura 1. Topología lógica del laboratorio.



4. Plan de direccionamiento

Equipo	Interfaz	IP	Máscara	Función
User	VPC	10.229.3.11	/25	Cliente DHCP
R1	Fa0/1	10.229.3.1	/25	Gateway Usuarios
R1	Fa0/0	203.0.113.2	/30	WAN
ISP	Fa0/0	203.0.113.1	/30	Tránsito público
FGT-2	port1	203.0.113.6	/30	WAN
FGT-2	port2	10.229.3.129	/28	Gateway Server
Server	eth0	10.229.3.130	/28	Web HTTPS

Redes: Usuarios 10.229.3.0/25; Servidores 10.229.3.128/28; R1-ISP 203.0.113.0/30; ISP-FGT-2 203.0.113.4/30.

5. Componentes

- Cisco C2691 — R1: gateway de usuarios y extremo IPsec.
- FortiGate — FGT-2: firewall, gateway del servidor y extremo IPsec; configurado por GUI.
- ISP: tránsito entre los extremos públicos.
- User: VPC con DHCP.
- Server: Ubuntu mínimo con Nginx y HTTPS.

6. Configuración de red

6.1 R1 — Cisco C2691

Fa0/0 = 203.0.113.2/30; Fa0/1 = 10.229.3.1/25. Ruta hacia 10.229.3.128/28 mediante 203.0.113.1.

```
R1#show ip interface brief
Interface                                IP-Address      OK? Method Status
      Protocol
FastEthernet0/0                        203.0.113.2     YES manual up
      up
FastEthernet0/1                        10.229.3.1      YES manual up
      up
NVI0                                    unassigned      NO  unset  up
      up
R1#
```

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter a
rea
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external typ
e 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-
IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user
static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

    203.0.113.0/30 is subnetted, 1 subnets
C       203.0.113.0 is directly connected, FastEthernet0/0
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       10.229.3.128/28 [1/0] via 203.0.113.1
C       10.229.3.0/25 is directly connected, FastEthernet0/1
S*    0.0.0.0/0 [1/0] via 203.0.113.1
R1#
```

6.2 Usuarios y DHCP

El User obtuvo 10.229.3.11/25 mediante DHCP, con gateway 10.229.3.1.

```
User> show ip
```

```
NAME       : User[1]  
IP/MASK    : 10.229.3.11/25  
GATEWAY    : 10.229.3.1  
DNS        : 8.8.8.8  
DHCP SERVER : 10.229.3.1  
DHCP LEASE  : 79646, 86400/43200/75600  
MAC        : 00:50:79:66:68:00  
LPORT      : 10055  
RHOST:PORT  : 127.0.0.1:10056  
MTU        : 1500
```

```
User>
```

```
User> ping 10.229.3.1
```

```
84 bytes from 10.229.3.1 icmp_seq=1 ttl=255 time=29.882 ms  
84 bytes from 10.229.3.1 icmp_seq=2 ttl=255 time=16.637 ms  
84 bytes from 10.229.3.1 icmp_seq=3 ttl=255 time=17.838 ms  
84 bytes from 10.229.3.1 icmp_seq=4 ttl=255 time=7.261 ms  
84 bytes from 10.229.3.1 icmp_seq=5 ttl=255 time=18.309 ms
```

```
User> 
```

7. NAT

R1 utiliza NAT overload para salida general. Se creó una excepción para que el tráfico VPN hacia 10.229.3.128/28 no sea traducido.

ACL NAT-INTERNET: deny 10.229.3.0/25 → 10.229.3.128/28; permit 10.229.3.0/25 → any.

```
R1#show ip nat translations

R1#show ip nat statistics
Total active translations: 0 (0 static, 0 dynamic; 0 extended)
Outside interfaces:
  FastEthernet0/0
Inside interfaces:
  FastEthernet0/1
Hits: 5 Misses: 45
CEF Translated packets: 45, CEF Punted packets: 10
Expired translations: 45
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-INTERNET interface FastEthernet0/0 refcount 0
Appl doors: 0
Normal doors: 0
Queued Packets: 0
R1#
```

8. Servidor Web HTTPS

Server: 10.229.3.130/28; gateway 10.229.3.129. Nginx quedó escuchando en 80 y 443. La página HTTPS muestra el propósito del laboratorio y la dirección del servidor.

```
root@Server:/# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
17: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 02:42:d1:cd:24:00 brd ff:ff:ff:ff:ff:ff
    inet 10.229.3.130/28 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:d1ff:fe2d:2400/64 scope link
        valid_lft forever preferred_lft forever
root@Server:/# ip route
default via 10.229.3.129 dev eth0
10.229.3.128/28 dev eth0 proto kernel scope link src 10.229.3.130
root@Server:/#
```

```
root@Server:/# curl -k https://127.0.0.1
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <title>Servidor Web - Infraestructura 1</title>
</head>
<body>
  <h1>Servidor Web HTTPS</h1>
  <p>Servidor: 10.229.3.130</p>
  <p>Laboratorio Infraestructura 1</p>
  <p>Conexión mediante VPN Site-to-Site</p>
</body>
</html>
root@Server:/#
```

9. VPN Site-to-Site

El túnel conecta 10.229.3.0/25 (R1) con 10.229.3.128/28 (FGT-2). Peers públicos: R1 203.0.113.2 y FGT-2 203.0.113.6.

- IKEv1, Main Mode, Pre-shared Key.
- PSK: VPN123.
- Phase 1: DES + SHA1, DH Group 5.
- Phase 2: DES + SHA1, PFS desactivado.
- Selectores: 10.229.3.0/25 ↔ 10.229.3.128/28.

FGT-2										
- Dashboard - Network - Policy & Objects - Firewall Policy - IPv4 DoS Policy - Addresses - Internet Service Database - Services - Schedules - Virtual IPs - IP Pools	Create New Edit Delete Policy Lookup Search Export Interface									
	Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
	port2 → port1									
	SERVER_TO_ISP	all	all	always	ALL	ACCEPT	Enabled	SSL no-inspection	UTM	672 B
	port2 → VPN-R1-FGT2									
	vpn_VPN-R1-FGT2_loca...	VPN-R1-FGT2_local	VPN-R1-FGT2_remote	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM	0 B
	vpn_VPN-R1-FGT2 → port2									
	vpn_VPN-R1-FGT2_rem...	VPN-R1-FGT2_remote	VPN-R1-FGT2_local	always	ALL	ACCEPT	Disabled	SSL no-inspection	UTM	840 B
	Implicit									

Phase 1 Proposal

[Edit](#)

Algorithms : DES-SHA1

Diffie-Hellman Group : 5

Phase 2 Selectors

Name	Local Address	Remote Address	Add	
VPN-R1-FGT2	VPN-R1-FGT2_local	VPN-R1-FGT2_remote		

```
R1#show crypto isakmp sa
dst          src          state          conn-id slot status
203.0.113.2  203.0.113.6  QM_IDLE        4           0  ACTIVE
```

```

R1#show crypto ipsec sa

interface: FastEthernet0/0
  Crypto map tag: VPN-MAP, local addr 203.0.113.2

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (10.229.3.0/255.255.255.128/0/0)
  remote ident (addr/mask/prot/port): (10.229.3.128/255.255.255.240/0/0)
  current_peer 203.0.113.6 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 19, #pkts encrypt: 19, #pkts digest: 19
    #pkts decaps: 19, #pkts decrypt: 19, #pkts verify: 19
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 6, #recv errors 0

```

10. Pruebas y resultados

10.1 VPN activa

Con el túnel activo, User pudo hacer ping a 10.229.3.130. R1 registró 9 paquetes encapsulados, 9 cifrados, 9 desencapsulados y 9 descifrados; la SA ESP DES/SHA quedó ACTIVE.

```

User> ping 10.229.3.130

84 bytes from 10.229.3.130 icmp_seq=1 ttl=62 time=97.290 ms
84 bytes from 10.229.3.130 icmp_seq=2 ttl=62 time=108.765 ms
84 bytes from 10.229.3.130 icmp_seq=3 ttl=62 time=77.946 ms
84 bytes from 10.229.3.130 icmp_seq=4 ttl=62 time=47.137 ms
84 bytes from 10.229.3.130 icmp_seq=5 ttl=62 time=77.515 ms

```

```

#pkts encaps: 19, #pkts encrypt: 19, #pkts digest: 19
#pkts decaps: 19, #pkts decrypt: 19, #pkts verify: 19
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 6, #recv errors 0

```

10.2 VPN desactivada

Se utilizó Bring Down → Entire Tunnel en FGT-2. Con el túnel apagado, el ping a 10.229.3.130 falló. Al volver a levantarlo, el ping volvió a funcionar. Esta prueba verifica el requisito principal.

VPN Tunnel Management			
Q >_ ? 3 admin + Create New Edit Delete Search Q			
Tunnel	Interface Binding	Status	Ref.
Custom 1			
VPN-R1-FGT2	port1	Inactive	4

```
User> ping 10.229.3.130

10.229.3.130 icmp_seq=1 timeout
10.229.3.130 icmp_seq=2 timeout
10.229.3.130 icmp_seq=3 timeout
10.229.3.130 icmp_seq=4 timeout
10.229.3.130 icmp_seq=5 timeout
```

VPN Tunnel Management			
Q >_ ? 3 admin + Create New Edit Delete Search Q			
Tunnel	Interface Binding	Status	Ref.
Custom 1			
VPN-R1-FGT2	port1	Up	4

```
User> ping 10.229.3.130

84 bytes from 10.229.3.130 icmp_seq=1 ttl=62 time=97.290 ms
84 bytes from 10.229.3.130 icmp_seq=2 ttl=62 time=108.765 ms
84 bytes from 10.229.3.130 icmp_seq=3 ttl=62 time=77.946 ms
84 bytes from 10.229.3.130 icmp_seq=4 ttl=62 time=47.137 ms
84 bytes from 10.229.3.130 icmp_seq=5 ttl=62 time=77.515 ms
```

11. Scripts utilizados

- R1-running-config.txt
- FGT-2-running-config.txt
- ISP-running-config.txt
- Server-network-config.txt
- Archivos/configuración de Nginx HTTPS, si fueron modificados.
- Cualquier script adicional utilizado durante el laboratorio.

12. Conclusión

La infraestructura cumplió el objetivo. El Usuario se comunicó con el Web Server mediante el túnel VPN Site-to-Site entre R1 y FGT-2. Los contadores IPsec confirmaron tráfico encapsulado y desencapsulado. Al desactivar el túnel, la comunicación dejó de funcionar; al reactivarlo, volvió a funcionar. La excepción de NAT evitó traducir el tráfico destinado a la red remota antes de entrar al túnel.